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Palaeomagnetism of Permian red beds from south Kazakhstan: DRM inclination error or CRM shallowed directions?

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SUMMARY

Thermal demagnetization of palaeomagnetic samples from Upper Permian red beds from south Kazakhstan yielded almost noiseless data. In seven samples that show rectilinear decay to the origin, the tilt-corrected mean direction of the characteristic component is $D = 217°$, $I = -44°$ ($K = 81$, $\alpha_{95} = 6.7°$). Most samples, however, reveal curved demagnetization paths and steady shift of vector endpoints on stereonet. For the latter samples, the NRM is accounted for by two components with similar declinations and different inclinations. The first tightly grouped component is an overprint and its mean inclination is close to the reference direction for stable Europe and to that for samples with rectilinear demagnetization plots. The second high-temperture component is more dispersed and about 15° shallower than the first one. Maximal K_{max} and intermediate K_{int} axes of anisotropy ellipsoids gravitate to the stereonet equator, whereas many K_{min} axes are vertical. Rock magnetic studies revealed haematite as the only remanence carrier in these rocks. We argue that the shallow inclination of the high-temperature component has not resulted from horizontal transport of the area, but is due to NRM acquisition process. One possibility is to assume that the most thermally stable component of detrital origin has inclination error and is overprinted by a component of chemical origin. Also, we discuss a model of time-distributed CRM acquisition and attribute the inclination shallowing to diagenetic compaction.

Key words: chemical remanent magnetization, inclination shallowing, red beds.

INTRODUCTION

One of the basic assumptions of palaeomagnetism is that the natural remanent magnetization (NRM) is acquired parallel to the direction of the ambient field. Some rocks, for instance, most lavas with thermoremanent magnetization, are believed to be faithful recorders of the ancient geomagnetic field. In contrast, the NRM in sediments may be shallowed due either to inclination error or compaction. The fact that compaction may bias palaeomagnetic directions is well established by both the results on natural sediments (Morgan 1979; Kent & Spariosu 1982; Tauxe *et al* 1984) and artificial compounds (Blow & Hamilton 1978; Noel 1980; Sun & Kodama 1992). These errors, however, are thought to be characteristic only for the detrital remanent magnetization (DRM); hence, secondary components are to be parallel to the direction of the ambient field during acquisition. A good example is red beds with a characteristic component of chemical origin (CRM) acquired after sedimentation. As a compensation for many problems connected with CRM data, mostly their dating, this remanence is thought to be directionally unbiased.

Is the latter statement really correct? Below we present palaeomagnetic results for Upper Permian red beds from south Kazakhstan which were sampled as a part of a tectonic project, but the revealed pattern is interesting enough to discuss it from purely palaeomagnetic point of view.

† Deceased 1994 March 8.

GEOLOGICAL SETTING AND SAMPLING

The area under study is close to WNW–ESE-striking early Palaeozoic (Caledonian) foldbelt of the Great Karatau Range where strongly deformed and often metamorphosed rocks are exposed. The early Palaeozoic basement under the northern foothills of this range is covered with a weakly deformed sequence of upper Palaeozoic age. The uppermost

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member of this sequence consists of lagoonal and lacustine intercalating siltstones, limestones and evaporites of the Tuzkul Formation with almost uniform northerly dips of no more than 15° (Fig. 1). Red to brown fine-graned sandstones and siltstone of this formation show diffuse vertical variations of colouration on a millimetre scale; small-scale cross-bedding is often present. Miospore assemblages from thin layers of black mudstones include *Protohaploxypinus perfectus Naum., Florinites luberas Samojl., Cordaitina urabensis Zub.*, and others (determined by L. N. Rzhannikova) and point to the Kungurian–Ufimian age of the formation. The Upper Paleozoic rocks are unconformably overlain by flat-lying Palaeogene sediments.

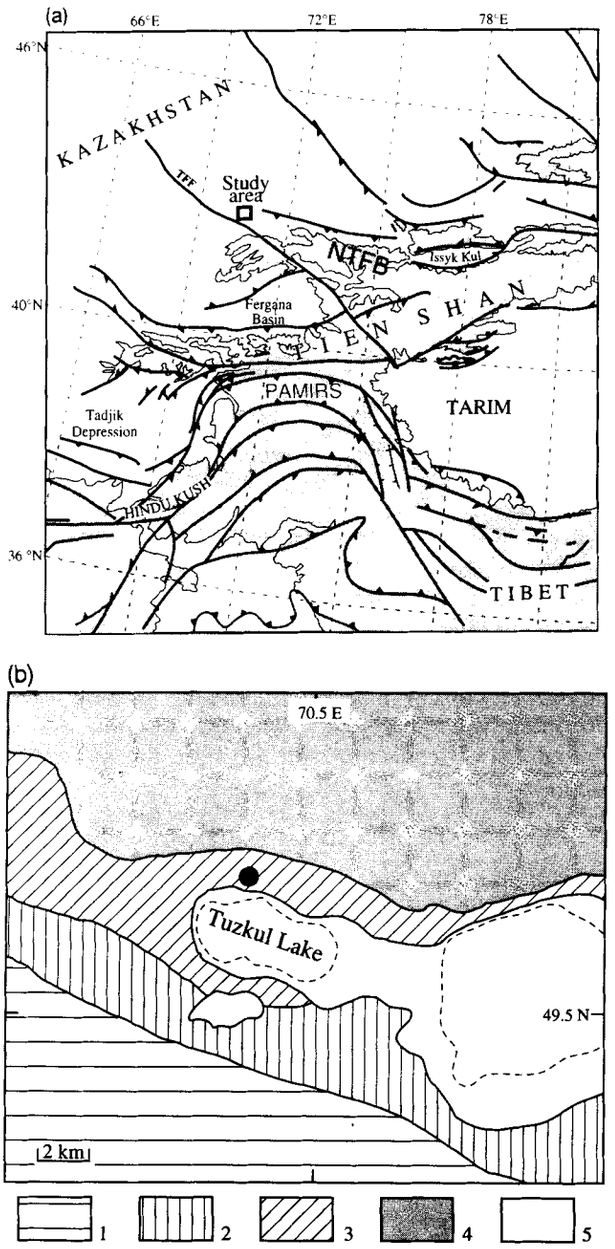


Figure 1. (a) Schematic map of Central Asia with elevations above 2,000 m shaded, and major faults shown as thick solid lines (TFF is the Talas–Fergana Fault) (after Cobbold & Davy 1988, simplified). (b) Geological map of the studies area with the sampled section as the filled circle. Legend: (1) Carboniferous; (2) Lower Permian; (3) Upper Permian (Tuzkul Formation); (4) Palaeogene; (5) Quaternary; the shorelines of seasonal salt lakes are shown as dashed lines.

The complete section of the Tuzkul formation known from boreholes is up to 700 m thick, but only its limited parts can be found in outcrops. From an outcrop on the northern shore of the Tuzkul Lake (Fig. 1b) we took 22 hand samples over about 100 m stratigraphic interval (20 samples of pale red to brown siltstones, one of red limestones and one of brick-red coarse-grained sandstone); samples were oriented with a magnetic compass. The direction of dip of the layers ranges from 5° to 15° towards the north and north-east.

LABORATORY PROCEDURES

The collection was studied in the palaeomagnetic laboratory of the Institute of Geology, Rennes University, France. Two cubic specimens from a hand sample were heated in up to 15 steps using a Schonstedt TSD-1 demagnetizer. Measurements were made on a LETI cryogenic magnetometer with the noise level at about 1×10^{-5} A m^{-1}. Components isolated from two specimens from a sample were used for calculations of sample means, which in turn were used for computation of the formation mean. Anisotropy was measured on one or two specimens from a sample with a Digico magnetometer. Other studies included chemical analyses, optic observations on thin sections, scanning electron microscopy and rock-magnetic experiments.

RESULTS

The NRM intensities of the red beds range from 2 to 10, about 4×10^{-2} A m^{-1} on average. A sample of brick-red coarse-grained porous sandstone completely remagnetized by the present-day field is rejected. After removal of a weak dispersed remanence at 200° to 250 °C, practically noiseless orthogonal plots are obtained for 21 remaining samples. Their inspection reveals that the plots vary from practically rectilinear (Figs 2a and b) through slightly curved (Figs 2c and d) to rather curved (Figs 2e and f); all vector endpoints are projected onto the upper hemisphere. (The curvature of demagnetization plots is better seen if a plot for each specimen is drawn in a coordinate system the axes of which are the eigenvectors of the vector endpoints distribution. In this case, however, it will be much more difficult to compare different specimens; hence, we preferred the common presentation). On stereonets, vector endpoints of some specimens are completely stable; the measured directions in many specimens, however, steadily shift along great circle segments. Though short, these segments are well defined (Fig. 3a), the normals to remagnetization circles fall within a narrow band on stereonet (Fig. 3b), and the mean direction of the least-dispersed component (Halls 1976) is precisely determined (Table 1). It is worth noting that vectors of all specimens move away from the least-dispersed component which is thus the least thermally stable. There is no correlation between lithology and demagnetization properties.

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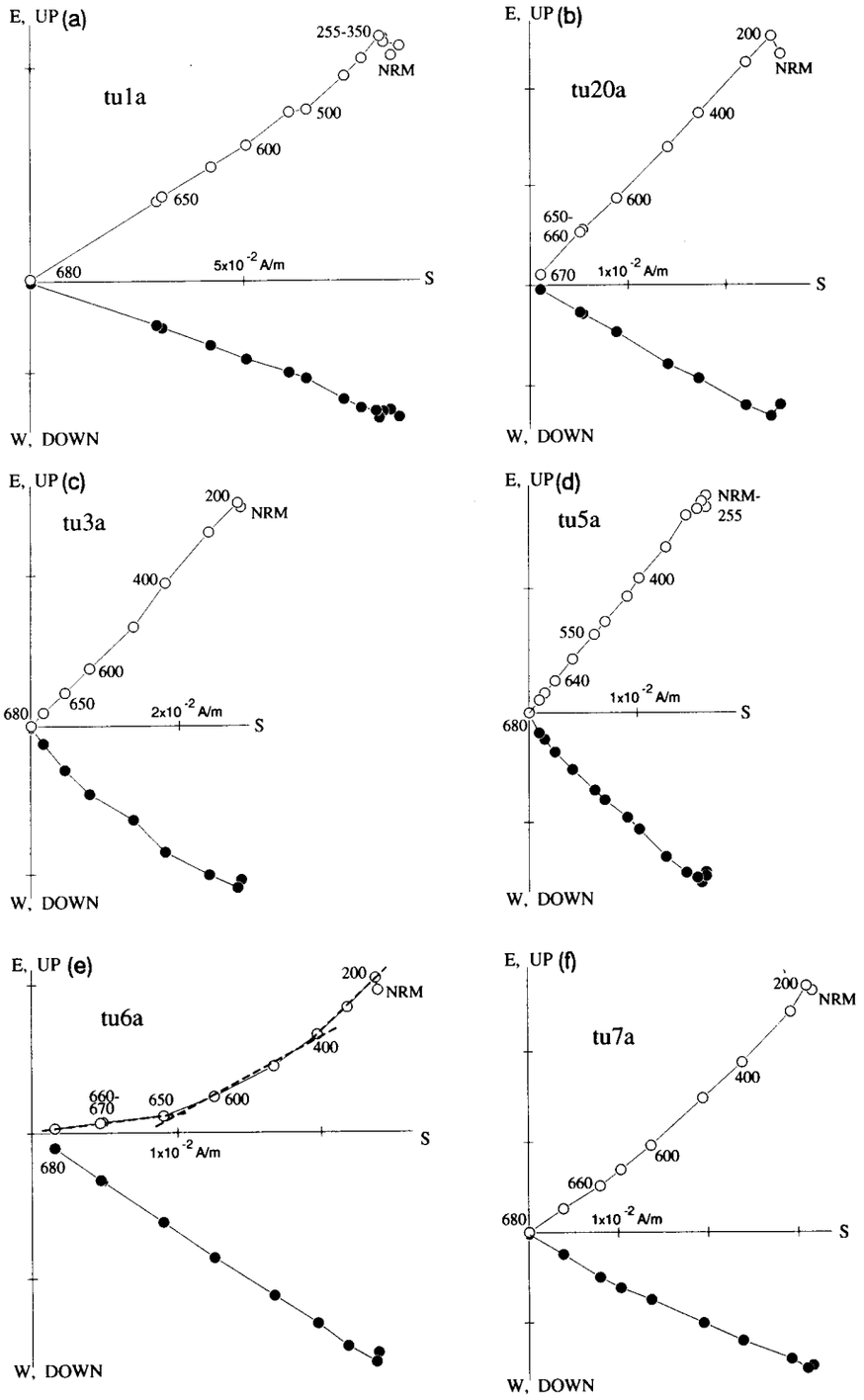


Figure 2. Representative orthogonal plots of the Tuzkul Formation. All projections are for the tilt-corrected data. Solid (open) symbols are projections onto horizontal (vertical) plane. Temperature is in degrees Celsius.

To illustrate the steady shift of vector endpoints for the entire collection, it is possible to calculate mean vectors for each demagnetization step, but we used another approach. All demagnetization data were divided into three windows, namely, 200° to 450°, 450° to 600° or 640°, and the final demagnetization steps. Then, linear segments were adjusted in each window, and the corresponding directions (LTD, ITD and HTD, respectively) were calculated for all specimens. Because of the smoothness of the orthogonal plots, tight fit of linear segments is possible in each window, mean angular deviations being about 2° on average. In a specimen, either inclination or declination or both may change during thermal demagnetization, but, for the entire data set, means declination remains the same over the entire

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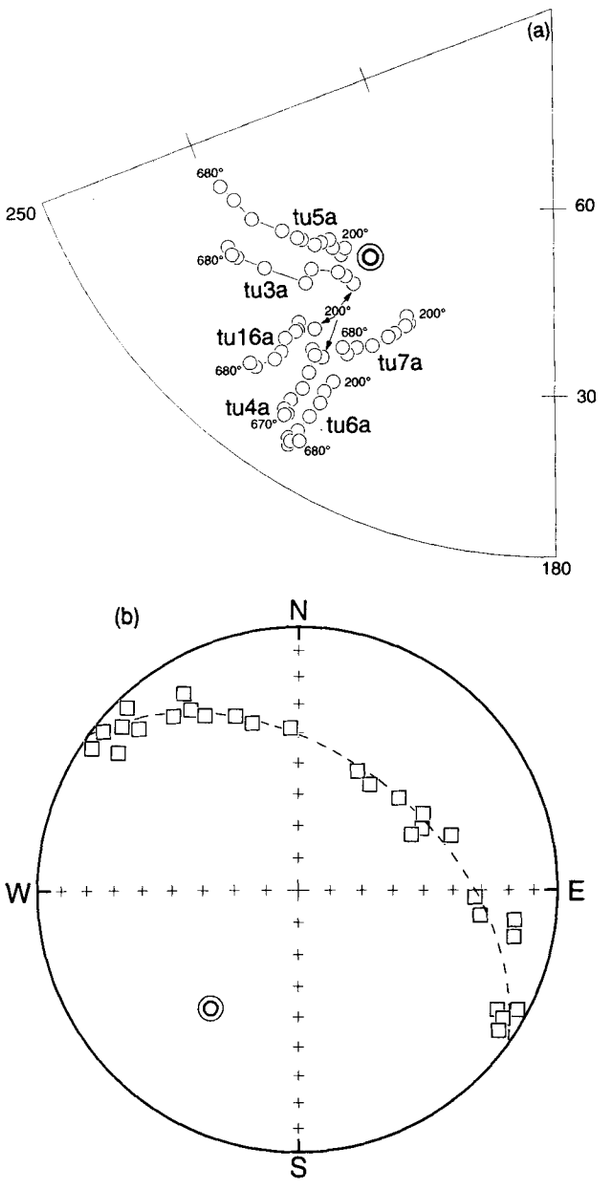


Figure 3. (a) Equal-area projection of (a) representative specimen vector-ends (dots) and (b) distribution of normals to remagnetization circles (squares) and the position of the best-fitting plane (line). All symbols and lines are projected onto the upper hemisphere; a double circle is the direction of the least-dispersed component determined as the normal to the best-fitting plane (Halls 1976).

temperature range, while the mean inclination steadily becomes shallower, and the scatter grows. It is also worth noting that the HTD distribution is elongated, the declination dispersion being larger than that of inclination, whereas the LTDs have a circular distribution.

Next, the collection was divided into two groups. The first group includes specimens that are directionally stable and show no movement of vector endpoint on stereonet, whereas the second group includes those that show consistent shift of directions. After this division mean directions for temperature windows were recomputed. As expected, the window means for the first group are essentially the same, whereas the differences in inclinations and dispersion between the window means become even more pronounced for the second group. Moreover, the overall mean for the first group coincides with the LTD mean for the second group and the result obtained by the converging great circles (Table 1).

Bedding attitudes of the sampled strata are rather uniform, and tilt correction leads just to a few per cent increase in precision parameter, and the fold tests (McElhinny 1964; McFadden & Jones 1981) are inconclusive.

Anisotropy measurements on 32 specimens from 16 samples reveal a rather distinct pattern (Fig. 4). Maximal and intermediate anisotropy coefficients are almost equal, and the anisotropy values range from 2 to 15 per cent, with the only exception of sample tu6 with anisotropy about 35 per cent. Two clusters can be seen on the plot of K_{max}/K_{int} versus K_{int}/K_{min}, one below 3 and the second between 6 and 10 values of the latter ratio (Fig. 4a). All the samples of the second cluster showed well-defined remagnetization circles during thermal cleaning; however, some specimens from the first cluster also behave similarly, e.g. sample tu16a (Fig. 3a) with anisotropy about 3 per cent. Most (but not all) specimens from the second cluster have vertical K_{min} axes, whereas K_{min} axes from most (but again not all) specimens from the first cluster fall into a more diffuse cloud. Maximal K_{max} and intermediate K_{int} axes of anisotropy ellipsoids gravitate to the stereonet equator (Fig. 4). Some preferences in K_{max} and K_{int} orientation are noticeable, but, taking into account the almost perfect equality of the two parameters, this is hardly significant. Unfortunately, a more distinct correlation is not found between palaeomagnetic properties, e.g. the length of the great circle segment, and K_{min} orientation or degree of anisotropy.

Rock-magnetic studies revealed high uniformity of the collection. All isothermal remanent magnetization (IRM) curves are very similar (Fig. 5a), and no correlation between directions of the NRM and IRM behaviour was found. Thermal demagnetization of the saturation magnetization also reveals rather uniform magnetic properties (Fig. 5b). In the latter figure, weak irregularities are noticeable on curves for samples tu6 and tu20, but they are not correlated with palaeomagnetic data: samples tu6 and tu7 have the strongest anisotropy and well-defined remagnetization circles, whereas samples tu1 and tu20 have very weak anisotropy and no shift of vector endpoints on stereonet.

Chemical analyses revealed that the main portion of iron resides in a phase that dissolves in concentrated hydrochloride acid, whereas the part of iron dissolved in weak acid is several-fold less; the share of iron in siderite and silicates is negligible. Thus, these analyses together with rock-magnetic and palaeomagnetic data point to haematite as the main iron compound and the only NRM carrier in these rocks.

Optical observations show that opaque minerals in these red beds are of two generations. Detrital grains often distinctly concentrate to boundaries between thin layers; the cross-bedded parts with high concentration of opaque minerals often display a one-sided feather-like pattern. In

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Table 1. Palaeomagnetic results.

		In Situ		Tilt Corrected			
TW	N	D°	I°	D°	I°	k	α_{95}°
Samples with unicomponent magnetization							
200-680	7	215	-50	217	-44	81	6.7
Samples with remagnetization circles							
LT	14	215	-48	215	-42	104	3.9
IT	14	213	-42	214	-36	77	4.6
HT	14	216	-34	217	-28	29	7.5
Least-dispersed component after Halls (1976)							
	(28)	-	-	214	-43	-	4.0

Comments. TW is temperature window: LT, IT and HT are 200°–450°, 450–640° and 640–680°, respectively (see text for explanation); N is the number of samples (specimens); D is declination; I is inclination; k is precision parameter (Fisher 1953); α_{95}° is the radius of confidence circle.

contrast, red pigmentation bands unconformably cut the sedimentation pattern and are thus of a later diagenetic origin. Scanning electron microscopy of representative specimens shows single angular grains of a Fe-Ti-rich mineral, presumably ilmenite, and, much less frequent angular grains of an iron oxide, presumably haematite: the latter, however, are very scarce and too large (about 50 μm in dimensions) to be oriented by the ambient field and thus carry a noticeable NRM component. No preferred orientation of silicate minerals was observed.

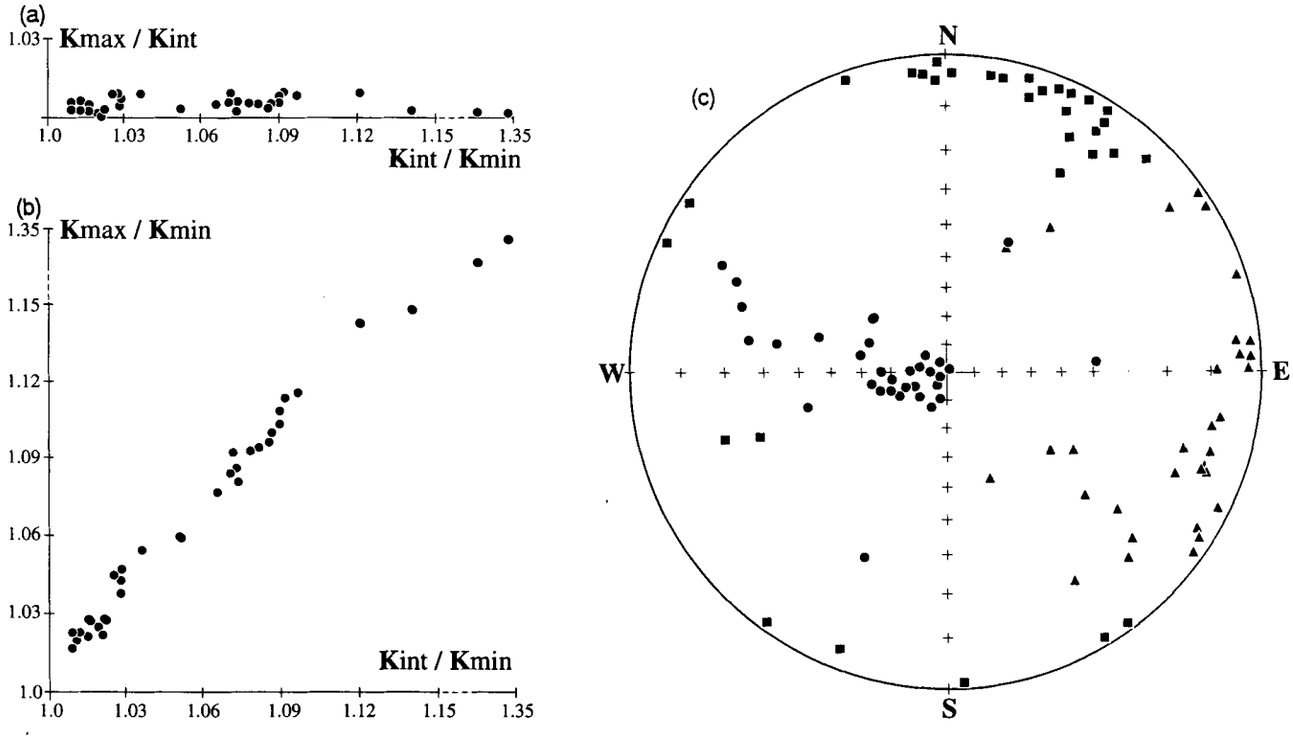


Figure 4. Anisotropy data on the specimen level. Plots of anisotropy coefficients (a, b) and distribution of maximal (squares), intermediate (triangles) and minimal (dots) axes of anisotropy ellipsoids after tilt correction (b).

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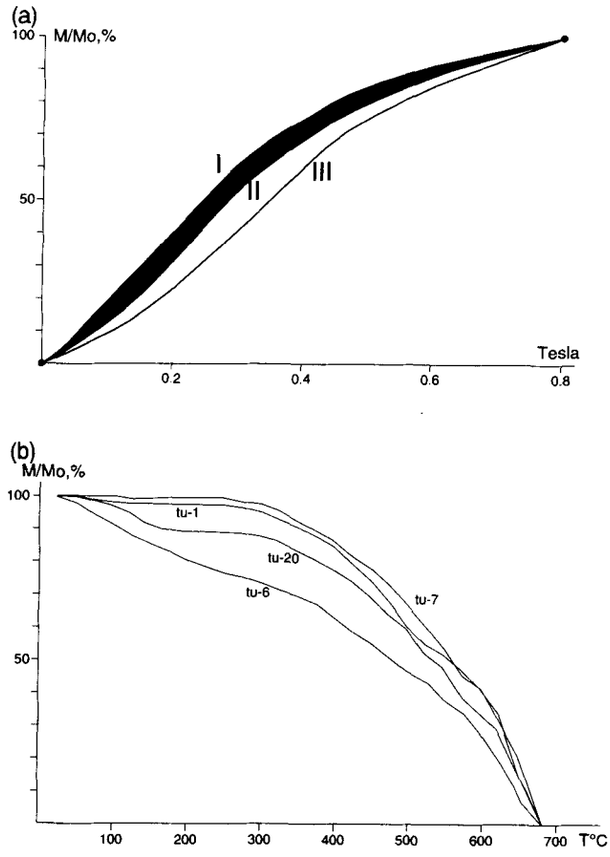


Figure 5. (a) Saturation IRM acquisition curves: I, II, two samples of Permian red siltstones (the curves for the other samples fall into the shaded area); III, the only sample of red limestone. (b) Thermal demagnetization of saturation magnetization for representative samples. For better comparison, paramagnetic 'tail' at temperatures above 680° is subtracted from each curve.

INTERPRETATION AND DISCUSSION

We think that, apart from the weak dispersed low-temperature component, the main part of the NRM in the Permian red beds may be accounted for by two components. A close similarity between the overall mean for the specimens with rectilinear orthogonal plots, the LTD mean for the specimens with curved plots and the result obtained by the converging great circles (Table 1) implies that the same remanence (component A) was found. Another component (B) should be present in these rocks to account for curved plots (Figs 2C–D) and remagnetization circles (Fig. 3A). The demagnetization pattern is thus the result of strongly overlapping unblocking spectra of these two components. Since all the measured directions and isolated components are of reversed polarity we think that the Permian red beds had acquired their magnetization during the Kiaman epoch.

It is questionable, however, whether the second component was properly isolated, and apparent stable endpoints may be due to complete overlap of unblocking spectra. Thus we cannot be sure that the HTD (Table 1) directly correspond to component B. Judging by our results, it is, however, reasonable to assume that a true direction of component B should have a declination similar to and an inclination not steeper than that for the HTD data. Keeping this in mind, we shall use the HTD mean as an estimation of component B further on.

For the area under study, the Eurasian palaeomagnetic poles (Van der Voo 1990) predict a steady shift of inclinations and nearly constant declinations since the Late Devonian to Late Permian (Fig. 6). The mean direction of component B differs by $20° \pm 6°$ from the Early Permian reference direction ($I = 47°$); in contrast, component A agrees well with the reference data. Taken at face value, shallow inclination of component B infers an about 1,500 km northward transport of the Tuzkul area with respect to Eurasia; even larger values are obtained if the Late Permian pole is used. However, there are no structures of appropriate age to the north, north-east and north-west that could have accommodated a displacement of this magnitude within Eurasia; Permian rocks there are either weakly folded and faulted or not deformed altogether. The general consensus is that Kazakhstan has already been welded to Eurasia by the mid-Permian (Khain 1979; Kharmov 1987). Early and Late Permian palaeomagnetic data from Kazakhstan (Kumpan, Rusinov & Sholpo 1968) agree well with the Eurasian reference data. Finally, Early Permian

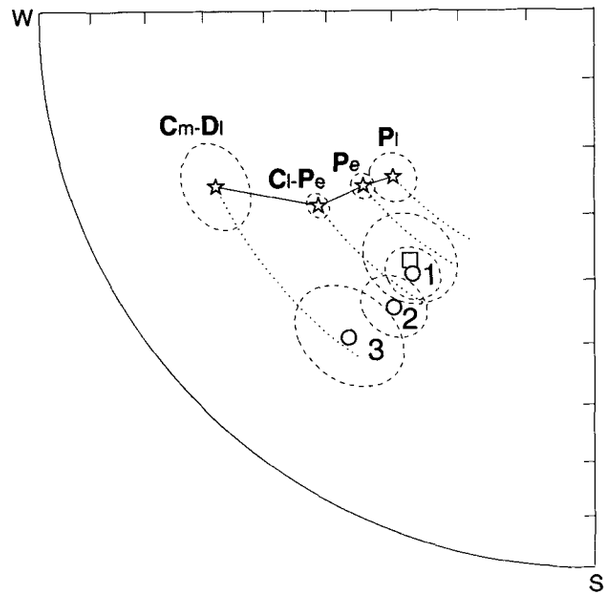


Figure 6. Equal-area projection of mean palaeomagnetic vectors and reference directions together with confidence circles. Dots are mean directions of components for each temperature window for the samples displaying remagnetization circles on stereonet: (1) 200°–450°; (2) 450° to 600° or 640°; (3) final steps (see text for explanations). Square is mean direction for the samples without distinct remagnetization circles. Stars are the European reference directions recalculated from mean poles (Van der Voo 1990): Pl, Late Permian (246–266 Ma); Pe, Early Permian (267–281 Ma); Cl–Pe, Late Carboniferous–Early Permian (282–308 Ma); Cm–Dl, Middle Carboniferous–Late Devonian (309–365 Ma). For convenience, mean reference inclination are also shown as dotted lines.

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results from the North Tien Shan fold belt that belongs to the same tectonic zone as the area under study, also closely fit the Eurasian reference data (Audibert & Bazhenov 1992). We thus conclude that the difference between the measured and reference inclinations is not due to tectonic movements, and palaeomagnetic vectors are directionally biased. It is worth noting that measured declinations do deflect westward with respect to the European reference data (Fig. 6); a possible rotation will be addressed in more detail elsewhere.

The common way to account for shallow inclinations in sediments is to assume that it is an inclination error during sedimentation or the result of compaction which rotated long axes of magnetic particles closer to horizontal plane. This implies that component B is the DRM, whereas component A is an overprint most probably of chemical origin. Also accounted for is the steady decrease of concentration parameter, since the CRM is usually much better clustered than the DRM.

This model, however, requires a close similarity of the two components, both residing in haematite of quite different origin. Optical and SEM data do reveal at least two generations of Fe-rich minerals; single large grains of detrital origin, however, can hardly be NRM carriers, let alone account for rather high magnetization intensities even well above 600°. At the same time, rock-magnetic and palaeomagnetic data point to high homogeneity of these rocks with respect to their magnetic carriers.

We assume that, apart from the weak remanence removed after heating at 200° or 250°, the entire NRM resides in haematite of chemical origin. This accounts for uniformity of palaeomagnetic and rock-magnetic properties. If this assumption is correct, shallowed inclinations are not due to inclination error, since the NRM is not detrital, but they result from compaction that is quite likely in these fine-grained rocks. Very distinct anisotropy pattern also favours compaction.

Laboratory studies show that inclination shallowing during compaction may be due to attachment of tiny magnetic debris to clay particles and/or randomization of these debris (e.g. Sun & Kodama 1992). It is well known that a re-alignment of non-isometric particles such as phyllosilicates takes place during compaction. Also known is the fact that the NRM in red beds is often carried by haematite coating on non-magnetic grains, as is most probably the case in the Permian rocks. We hypothesize that compaction and acquisition of the CRM were synchronous, and inclination shallowing occurs because of re-alignment of non-isometric haematite-coated particles. It should be repeated here that no preferred alignment of silicates was revealed in our case by SEM observations. The Permian red beds, however, are very heterogeneous with respect to their granulometry, grains varying from siltstone to clay fraction, whereas the experiments on compaction-induced shallowing were performed on pure clay minerals (Sun & Kodama 1992).

How to account for the fact that the particles that carry the shallowed HTC are the most thermally stable ones? Actually, the sediment reddening and related CRM acquisition are long-term processes. One may assume that the initial part of the NRM was acquired rather early during diagenesis. As compaction and CRM acquisition went on 'side by side', magnetization of newly formed haematite particles was less and less distorted. The haematite particles that had appeared first are also the largest and thus are the most thermally stable, whereas unblocking temperatures are less for finer grains. It was shown that a more or less uniform inclination shallowing, irrespective of its origin, leads to a growth of dispersion (Bazhenov 1981). Really, if f is nearly constant in $\tan I_{mes} = f \tan I_0$, where I_{mes} and I_0 are measured and original inclinations; respectively (King 1955), the dispersion of I_{mes} alone will be almost the same as for I_0, declinations will remain the same, but the dispersion of unit vectors will increase, and they will distributed banana like along a small circle. A similar growth of dispersion is really observed (Table 1).

So far, we have worked on an assumption that these red beds accumulated on a horizontal surface, and their NRM predates the tilt of strata. If the first assumption is incorrect, and these rocks accumulated on a tilted surface, the entire above discussion remains valid. Moreover, the *in situ* mean directions for the 'rectilinear' subset and 200°–400° temperature window (Table 1) are in perfect agreement with the Permian reference data, whereas the same mean directions after tilt correction are slightly shallower (Fig. 6). Nevertheless, the primary tilt hypothesis does not seem to be acceptable as the accumulation of the several-hundred-metre thick series of fine-grained clastics, limestones and evaporites on the slope with the tilt about 8° is unlikely. In contrast, if the second assumption is false, and the remanence acquisition postdates the tilting, the only way to account for the observed pattern is to assume the CRM shallowing.

The Permian red beds are not the only object where shallowing of CRM directions may be suspected. An overprint of Upper Cretaceous age was found both in red sandstones and mudstones of the Lower Cretaceous age from the Fergana basin (Bazhenov 1993). This overprint resides in haematite, and its secondary nature was deduced from orthogonal plots and negative conglomerate test performed on an intraformational conglomerate. At the same time, the mean inclination in the sandstones is steeper by about 12° than in the mudstones. This finding was tentatively attributed to compaction; one of the reviewers of this paper, however, mentioned that CRM shallowing has never been reported.

We do not imply that our results and reasoning have proved beyond any doubt that CRM directions are shallowed, and this hypothesis should be additionally tested on the other objects. If proven to exist, CRM shallowing may require some changes in palaeomagnetic work, first of all a more sophisticated sampling of various lithologies. On the other hand, this phenomenon may in some cases help to date the CRM data more precisely, since compaction usually takes place at early stages of diagenesis.

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